What Item Response Theory Adds to a Questionnaire: A Psychometric Analysis of the DASS-21

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Abstract

This essay has two parts. The first is a reflective account of how my understanding of item response theory (IRT) developed over the course, organised around the shift from a theory of *tests* to a theory of *items*. The second analyses the 21-item Depression Anxiety Stress Scales (DASS-21) in a sample of 1222 respondents with classical test theory (CTT), binary IRT, differential item functioning (DIF), categorical confirmatory factor analysis (CFA), measurement invariance and a graded response model. The methods converge on one picture. The scale is dominated by a strong general factor of negative affect (ω_H = .908, ECV = .808); the three subscales correlate at .79–.85 at the latent level and keep little reliable variance of their own once the general factor is controlled (ω_S = .081 to .224). A total score is defensible, separate subscale scores much less so. No measurement model fits exactly: the one-factor model is clearly rejected and even the best (bifactor) model fits only acceptably. Items function slightly differently across the two gender groups, in both directions and with negligible effect sizes; thresholds are approximately, not exactly, invariant, and latent mean comparisons are robust to freeing the least invariant items. Dichotomising the four-point items, as the exercise requires, leaves the graded response model with 2.43 times as much test information and changes which items look biased – the clearest argument in these data for keeping the original response format.

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1 Introduction

The DASS-21 was built to separate depression, anxiety and stress (P. F. Lovibond & Lovibond, 1995). Whether it succeeds is an empirical question that different psychometric frameworks answer in different ways, and the disagreements between those answers are more instructive than any single one. Classical test theory asks whether the total score is internally consistent; item response theory asks where on the latent continuum each item is informative; factor analysis asks how many latent variables are needed and whether a sum score is a defensible stand-in for any of them.

Section 2 is the reflective part required by the assignment. Section 3 describes the data and the analytic decisions, in particular the treatment of missing data and the dichotomisation of four-category items. Section 4 reports the results, Section 5 reads them across methods – where CTT, IRT and CFA agree, where they disagree and why – and Sections 6 and 7 state what the data cannot support.

Two framing points matter from the start. The dataset comes without a codebook, so the two gender codes are reported as "Group 1" and "Group 2" and their substantive labels are treated as an inference (Section 3.2). And the assignment asks for a binary IRT analysis of items that were not designed to be binary; I treat that requirement as a measurement question in its own right and quantify what it costs (Section 4.8).

2 A Reflective Account of Item Response Theory

2.1 What I expected IRT to be, and what it turned out to be

Before the course I understood IRT as a more sophisticated alternative to summing item scores: a family of logistic models that produce a better ability estimate than a raw total. That description is not wrong, but it is a description of the output rather than of the idea, and it misses what actually distinguishes IRT from what came before. The reframing that changed my reading of the whole subject is the one made explicitly in the lecture on classical test theory: CTT is a theory of *tests*, IRT is a theory of *items* (Hambleton & Swaminathan, 1985; Lord & Novick, 1968).

The consequence is easier to see through the CTT machinery. In CTT the observed score decomposes as $X = \tau + \epsilon$, and everything the theory can say is a statement about the variance of τ relative to the variance of X in a particular population, using a particular test. Reliability is a property of a test-in-a-population, not of a measurement procedure. Item difficulty is the proportion correct *in this sample*; item discrimination is a correlation *with this total score*. Both change if the sample changes, and neither licenses any statement about how a specific person would respond to a specific item. This is not a minor inconvenience. It means that the two things applied researchers most want to do – compare people who took different items, and compare items across different groups of people – are exactly the things CTT cannot underwrite.

IRT changes the object of modelling. Instead of modelling a total score, it models the probability of a response conditional on a person parameter and an item parameter, $P(X_{ij} = 1 \mid \theta_j, \beta_i)$. Once that conditional probability is the primitive, persons and items live on the same scale, and questions that were unanswerable in CTT become ordinary: how much information does this item provide at $\theta = 1$; is this item behaving differently for men and women who are equally depressed; what would the standard error of measurement be for a person at the top of the distribution. What I initially read as extra statistical apparatus turned out to be a change in what counts as the unit of analysis.

2.2 The Rasch model, specific objectivity, and why the model is restrictive on purpose

The Rasch model (Rasch, 1960) was the part of the course I misjudged most. My first reading was that it is the simplest IRT model, a 2PL with the discrimination parameters set to one for convenience, and therefore the model one uses when data are sparse. That reading inverts the logic. In the Rasch model the

constraint is the point. Because the model has the form

$$P(X_{ij} = 1 \mid \theta_j, \beta_i) = \frac{\exp(\theta_j - \beta_i)}{1 + \exp(\theta_j - \beta_i)}, \tag{1}$$

the log-odds of a correct response is a *difference* between a person parameter and an item parameter. Comparisons of two persons therefore do not depend on which items were used, and comparisons of two items do not depend on which persons responded – the property Rasch called specific objectivity. The raw score becomes a sufficient statistic for θ, which is why the item parameters can be estimated conditionally, eliminating the person parameters entirely rather than assuming a distribution for them (Fischer & Molenaar, 1995); the statistical foundations of this construction are set out by Holland (1990).

I found the practical consequence of this genuinely surprising: conditional maximum likelihood estimation of item difficulties requires no assumption about how θ is distributed in the sample. That is a strong property, and it is purchased by a strong restriction. The 2PL relaxes the restriction by letting each item have its own slope, and buys better fit at the cost of sufficiency – the raw score no longer carries all the information about θ, and the weighted score $\sum_i a_i x_i$ takes its place. This is the trade-off I now think is the most useful single thing to carry out of an IRT course. It is not a technical choice between two link functions; it is a choice about whether one wants a measurement instrument with strong invariance properties or a statistical model that describes the data well. The two goals genuinely conflict, and the Rasch/2PL debate is essentially a disagreement about which one should give way. The contrast is the one Embretson and Reise (2000) summarise as the "old" and the "new" rules of measurement.

2.3 Item and person parameters as locations, not coefficients

Item characteristic curves were the point at which the parameters stopped feeling like regression coefficients. The location parameter β_i is the value of θ at which a respondent has a 50% chance of endorsing the item; the discrimination a_i is the slope at that point. Reading them as *positions on the same ruler as the persons* rather than as effect sizes makes the person–item map (Figure 3) the natural summary of a test, and it makes the central design question obvious: are the items located where the people are? A test can be perfectly reliable in the CTT sense and still be badly targeted, because α averages over the whole sample and cannot reveal that the items all cluster in one region of the continuum.

The discrimination parameter took longer to interpret substantively. A steep item separates people sharply, but only in a narrow band around its location; a flat item separates them weakly but over a wider band. There is no such thing as a uniformly good item, only an item that is good for a purpose. That reframing – from "how good is this item" to "good where, and for whom" – is the practical lesson I expect to use most.

2.4 Information, and the end of a single reliability coefficient

The information function was the most useful formal idea in the course, and the one I would most want to explain to someone who has only met CTT. For the 2PL,

$$I_i(\theta) = a_i^2 \, P_i(\theta)\big[1 - P_i(\theta)\big], \qquad I(\theta) = \sum_i I_i(\theta), \qquad \mathrm{SE}(\hat{\theta}) = \frac{1}{\sqrt{I(\theta)}}. \tag{2}$$

Three things follow that CTT cannot express. First, information is additive across items, so the contribution of each item to the precision of the test can be inspected separately – item selection becomes a design problem rather than a matter of deleting whatever lowers α. Second, precision is a *function of* θ: a test can be excellent in the middle of the scale and nearly useless at the extremes, which is exactly what a single standard error of measurement hides. Third, in the Rasch model $P(1 - P)$ is simultaneously the conditional variance of the response and the item information, so the residual, the information function and the fit statistics are all expressions of the same quantity. When I saw that the infit and outfit mean

squares are just variance-weighted averages of squared standardised residuals $Z_{ij} = (X_{ij} - P_{ij})/\sqrt{P_{ij}Q_{ij}}$, several parts of the course that I had been treating as separate topics collapsed into one.

The empirical payoff appears in Section 4.8: once the four response categories are kept, the DASS-21 in this sample is most informative well *above* the mean of the trait and loses precision at the low end – something no reliability coefficient in Section 4.2 could have told me.

2.5 Assumptions: unidimensionality, local independence, monotonicity

Local independence is the assumption I understood least well at first and now regard as the substantive core of the model. Formally it states $P(\mathbf{X} = \mathbf{x} \mid \theta) = \prod_i P(X_i = x_i \mid \theta)$: conditional on the latent variable, responses are independent. What makes it substantive is that it is the operational definition of "the latent variable explains the associations among the items". Dimensionality and local dependence are therefore not two separate assumptions but two ways in which the same requirement can fail, which is why the lecture insists that general goodness-of-fit tests cannot distinguish them: a residual correlation between two items may reflect an extra dimension, or shared wording, or a response set, and the statistic is silent about which.

This is also where I remain least comfortable. Item pairs in the DASS-21 such as item 4 ("breathing difficulty") and item 15 ("close to panic") correlate more strongly than the model predicts (Section 4.5). Is that a fourth dimension, or is it two paraphrases of the same experience? The data cannot decide, and the honest position is that local dependence is diagnosed statistically but explained substantively – a division of labour that no amount of additional modelling removes.

2.6 The bridge to factor analysis: the same model in two notations

The lecture that changed my mental map most was the one showing that item factor analysis and IRT are the same model. Positing a latent continuous response $Y_i^* = \lambda_i \xi + \epsilon_i$ with a threshold τ_i, so that $Y_i = 1$ when $Y_i^* > \tau_i$, gives

$$P(Y_i = 1 \mid \xi) = \Phi\left(\frac{\lambda_i \xi - \tau_i}{\sqrt{\operatorname{var}(\epsilon_i)}}\right), \qquad a_i^{\text{probit}} = \frac{\lambda_i}{\sqrt{1 - \lambda_i^2}}, \qquad b_i = \frac{\tau_i}{\lambda_i}, \tag{3}$$

and multiplying a_i by $D = 1.702$ converts the normal-ogive parameters into the logistic metric of a 2PL. Tetrachoric and polychoric correlations are not a workaround for ordinal data; they *are* the IRT model, written as a correlation structure. The polychoric factor model with four categories is Samejima's graded response model.

The practical consequence is that the two literatures answer different questions about one object. Factor analysis is the better instrument for dimensionality, because it models the whole correlation structure and offers nested model comparison; IRT is the better instrument for individual measurement, because it puts persons and items on one scale and delivers conditional standard errors. In Section 5 I convert the CFA loadings into IRT discriminations and compare them with the directly estimated 2PL parameters, which is the clearest way I know of making the equivalence concrete rather than notational.

2.7 Reliability, dimensionality and why α is the wrong default

Before the course I would have reported Cronbach's α (Cronbach, 1951) and moved on. Three points changed that. First, α equals reliability only under essential τ-equivalence – equal loadings – and otherwise underestimates it; the model-based ω replaces an assumption with an estimate. Second, and more important, α is a function of the number of items and the average inter-item correlation only, and therefore says nothing about dimensionality. A high α is compatible with one homogeneous factor and with two barely related subtests. Third, once one accepts that a scale may be multidimensional, "reliability" stops

being a single number: the bifactor decomposition distinguishes ω_T (variance due to all common factors) from ω_H (variance due to the general factor alone), and their ratio answers the question that actually matters when someone reports a total score (Reise, Bonifay, & Haviland, 2013) – *what is that score a measure of?* The corresponding subscale coefficient ω_S answers the parallel question for subscale scores, and in this dataset it delivers the single most consequential result of the whole analysis (Section 4.6).

A further subtlety I would previously have ignored: for categorical indicators the usual ω is computed in the metric of the *latent* responses Y^*, not of the observed item scores. The Green and Yang (2009) correction re-expresses reliability in the observed metric using the thresholds and the bivariate normal integrals. The gap between the linear and the non-linear estimate is not an inconvenience to be reconciled; as the lecture puts it, its magnitude is an estimate of how badly the linearity and normality assumptions are violated.

2.8 DIF and measurement invariance: the same question, twice

DIF and measurement invariance were taught in different weeks, in different notations, and it took me some time to see that they ask one question: does the relationship between the latent variable and the response depend on group membership? A uniform DIF effect is a difference in item location, which in the factor-analytic model is a difference in thresholds; non-uniform DIF is a difference in discrimination, which is a difference in loadings. Once that correspondence is clear, the multi-group CFA sequence (configural, then thresholds, then loadings, then residuals) is simply an omnibus DIF test with a different accounting of degrees of freedom.

Two cautions from the lectures have stayed with me. The first is that DIF is a statistical finding, not a verdict of unfairness (Martinková et al., 2017): an item can function differently across groups for reasons that are substantively legitimate, and a content review is needed before an item is deleted. The second is that with $N > 1000$ the χ^2 difference test will reject almost any equality constraint, which is precisely why ΔCFI and ΔRMSEA criteria exist. Both cautions turn out to be decisive for this dataset, where the statistical and the practical verdicts point in opposite directions (Section 4.7).

2.9 What remains difficult, and where I would use this

Three things still feel unresolved to me. (i) *Model choice as a value judgement.* The Rasch model can be defended as a normative standard that data must meet, or rejected as a model that demonstrably does not fit; both positions are internally coherent and the data cannot arbitrate. (ii) *Fit statistics that test too much at once.* A significant χ^2, an inflated outfit or a large residual correlation can each arise from several distinct violations, and the statistics do not identify which. Fit assessment is therefore much more interpretive than its formalism suggests. (iii) *Factor score indeterminacy.* That infinitely many sets of factor scores are consistent with the same model (Bollen, 2002; Grice, 2001) is a genuinely uncomfortable fact, and it is the strongest argument I know for the otherwise unsophisticated sum score.

Where would I use this? The ideas that transfer most readily outside psychometrics are conditional reliability and DIF. Any predictive model built on a questionnaire, a rating scale or an annotator-labelled outcome inherits the measurement properties of that instrument, including the fact that its precision varies across the range of the construct and that it may behave differently across subgroups. A model evaluated only on aggregate accuracy cannot detect either. Framing that as measurement rather than as a modelling problem – and having the vocabulary of information functions and invariance to express it – is the part of this course I expect to keep using.

3 Data and Methods

3.1 The DASS-21

The DASS-21 is the short form of the Depression Anxiety Stress Scales (P. F. Lovibond & Lovibond, 1995; S. H. Lovibond & Lovibond, 1995). Respondents rate how much each statement applied to them over the past week, from 0 = "did not apply to me at all" to 3 = "applied to me very much, or most of the time". Seven items each form Depression (3, 5, 10, 13, 16, 17, 21), Anxiety (2, 4, 7, 9, 15, 19, 20) and Stress (1, 6, 8, 11, 12, 14, 18); no item is reverse keyed. Because the scale was designed to discriminate depressive from anxious symptoms, the overlap between its subscales is a substantive question rather than a technicality. Item content quoted below is that of the published English DASS-21; the dataset contains only numeric responses.

3.2 Sample and variables

The file contains $N = 1222$ respondents with gender, age, years of education and marital status. Age ranges from 18 to 78 years ($M = 43.38$, $SD = 14.05$), education from 5 to 30 years ($M = 13.87$, $SD = 3.73$). Gender is coded 1 ($n = 491$) and 2 ($n = 731$).

No codebook accompanies the data, and the two available clues about the gender codes conflict. The final character of the identifier is perfectly confounded with the code ("M" occurs only in group 2, "P" only in group 1), which suggests group 2 is male. The mean differences point the other way: group 2 scores higher on all three subscales ($d = .22$, .34 and .28 for Depression, Anxiety and Stress), the direction usually reported for women (Henry & Crawford, 2005; Osman et al., 2012). I therefore report "Group 1" and "Group 2"; every statistical conclusion is invariant to the labelling, only the substantive reading of group differences is not.

3.3 Missing data

Missing values are stored as Excel `#NULL!` error cells and are read as missing. 8 of the $1222 \times 21 = 25{,}662$ item responses are missing (0.03%); 8 respondents (0.7%) miss exactly one item each. Age and education have two and three missing values.

Listwise deletion is unbiased only under MCAR and loses power (Enders, 2010; Rubin, 1976); pairwise deletion (the lavaan default for categorical data) uses more information but lets the effective sample differ between correlations; full-information and multiple-imputation approaches add assumptions and Monte Carlo error. Here the choice has almost no empirical consequence: missingness is not associated with group (Fisher's exact test, $p = .28$), although with eight incomplete cases this test can only exclude a gross pattern and cannot establish MCAR. The eight incomplete cases score somewhat higher on the items they did answer, so they are not a random subset in every respect; but eight of 1222 respondents cannot move means, correlations or loadings appreciably. I use the complete cases ($N = 1214$; $n_1 = 486$, $n_2 = 728$) throughout, so that CTT, IRT, DIF, CFA, invariance and the graded response model are all estimated on the same respondents and differences between methods cannot be artefacts of differing samples. The two respondents without age and the three without education are additionally excluded from the analyses that use those variables.

3.4 Dichotomisation

The response distributions are strongly skewed (Figure 1): category 0 is the most frequent response for 10 of the 21 items and category 3 never exceeds 12.1% of responses. A per-item median split was rejected because the same binary code would mean different things for different items. Two interpretable rules remain, and both are carried through the binary analyses:

Rule A (0 vs. 1–3): "the symptom applied at all". It follows the anchor wording, which places the sharpest break between "not at all" and any endorsement, and gives endorsement rates from .31 to .89 ($\bar{p}$ = .57); only 2.1% of respondents endorse no item and 6.1% all 21. This is the primary analysis.

Rule B (0–1 vs. 2–3): "applied to a considerable degree". Closer to a clinical screening threshold, but endorsement rates fall to .10–.45 ($\bar{p}$ = .20) and 35.1% of respondents endorse no item. It serves as a sensitivity analysis.

Either rule discards the ordering within the collapsed categories (Cohen, 1983; MacCallum et al., 2002); the cost is quantified in Section 4.8.

3.5 Classical test theory

Item means, standard deviations, response distributions, corrected item–total correlations and α-if-item-deleted were computed for the total scale and each subscale; α is reported with a Feldt confidence interval (Feldt et al., 1987), together with the average inter-item correlation and the standard error of measurement. Because α presupposes essential τ-equivalence (Cronbach, 1951), model-based ω coefficients are reported alongside it (Bentler, 2009; McDonald, 1999; Raykov, 2001; Sijtsma, 2009).

3.6 Binary IRT

For each rule the Rasch model was estimated by conditional maximum likelihood (Mair & Hatzinger, 2007; Rasch, 1960), which needs no assumption about the distribution of θ; item fit was examined with infit and outfit mean squares (conventional acceptable range 0.7–1.3, Linacre, 2002; Wright and Linacre, 1994) using maximum-likelihood person parameters, and global fit with Andersen's conditional likelihood-ratio test for a median split of the raw score and for gender (Andersen, 1973). By marginal maximum likelihood with $\theta \sim N(0, 1)$ (Chalmers, 2012) I fitted the 1PL (one slope common to all items – the Rasch model in marginal form) and the 2PL (Birnbaum, 1968), compared them by a likelihood-ratio test and AIC/BIC, and assessed the 2PL with the limited-information M_2 statistic and the $S\text{-}X^2$ item-fit statistic (Maydeu-Olivares & Joe, 2006; Orlando & Thissen, 2000). Item and test information and the conditional standard error $1/\sqrt{I(\theta)}$ were computed on a grid from −4 to 4; marginal reliability is $1 - \overline{SE^2}/(\mathrm{var}(\hat{\theta}) + \overline{SE^2})$ using EAP estimates.

3.7 Differential item functioning

Three procedures implemented in `difR` (Magis et al., 2010) were used, each with iterative purification and matching on the total score. Mantel–Haenszel (Holland & Thayer, 1988) gives the ETS effect size $\Delta_{MH} = -2.35 \ln \alpha_{MH}$, classified as negligible (A), moderate (B) or large (C: $|\Delta_{MH}| \geq 1.5$ and significant) (Zieky, 1993). Logistic regression (Swaminathan & Rogers, 1990; Zumbo, 1999) tests uniform and non-uniform DIF, with Nagelkerke ΔR^2 as effect size and the Jodoin and Gierl (2001) classes (< .035 negligible, .035–.070 moderate). Lord's χ^2 compares separately calibrated 2PL parameters. The direction of each effect was fixed by the standardised p-difference (focal minus reference endorsement proportion at equal rest score), independently of package sign conventions. Multiplicity was controlled with Benjamini–Hochberg within each grouping variable. Gender is the required grouping variable. As bonus analyses I examined age and education, both plausible sources of differential symptom reporting, split at their medians for comparability with gender; because dichotomising a continuous covariate is the same practice this essay criticises for the items, the logistic-regression test was repeated with age and education as continuous covariates. Marital status was not used: its smallest category has eight respondents and it is strongly confounded with age (mean age 27 among single versus 49–60 in the other categories).

3.8 Confirmatory factor analysis

All CFA models use the original four-category responses as ordered indicators, with diagonally weighted least squares on the polychoric correlations and a mean- and variance-adjusted test statistic (WLSMV), delta parameterisation and latent variances fixed to one (Muthén, 1984; Olsson, 1979; Rosseel, 2012). Four models were fitted: (M1) one general factor; (M2) three correlated factors following the official subscales; (M3) a second-order factor above M2's three factors; (M4) a bifactor model with one general and three orthogonal specific factors. With three first-order factors M3 is just identified at the higher level and reproduces M2's fit exactly; it is kept because its loadings are informative. Fit is judged with the scaled χ^2, CFI, TLI, RMSEA and SRMR against the usual benchmarks (Hu & Bentler, 1999); nested models are compared with scaled difference tests (Satorra & Bentler, 2001, 2010) and ΔCFI (Cheung & Rensvold, 2002); residual polychoric correlations are inspected for local dependence (Yen, 1984). CFI and RMSEA are reported throughout in their *scaled* form; lavaan's "robust" versions are mentioned as a sensitivity check because they cannot be computed for the invariance models with constrained thresholds.

3.9 Reliability and dimensionality

All coefficients were computed from the standardised loadings with the formulas of the lecture: congeneric ω, ω_T, ω_H, ω_H/ω_T, explained common variance (ECV), percentage of uncontaminated correlations (PUC), subscale ω and ω_S, average variance extracted (Fornell & Larcker, 1981), construct replicability H and factor determinacy (Reise, 2012; Reise, Bonifay, & Haviland, 2013; Rodriguez et al., 2016). Each ω is computed twice: linearly, in the metric of the continuous latent responses, and with the Green and Yang (2009) correction, which re-expresses reliability in the metric of the observed 0–3 scores through the thresholds and bivariate-normal probabilities. `BifactorIndicesCalculator` (Dueber, 2020) served as an independent check, and an exploratory bifactor solution (Schmid–Leiman, `psych::omega`, Revelle, 2024) as a third route to ω_H. Benchmarks are those of the lecture: $\omega_H > .70$, ECV $> .60$ acceptable and $> .70$ good, PUC $> .80$ as the point above which the other indices matter less; determinacy is judged against .80–.90 (Ferrando & Lorenzo-Seva, 2018; Gorsuch, 1983).

3.10 Measurement invariance

Invariance across groups was tested by multi-group CFA on the four-category data (Little, 1997; Meredith, 1993; Millsap & Yun-Tein, 2004), in the sequence configural → metric (equal loadings) → scalar (equal loadings and thresholds; for ordinal indicators the thresholds take the role of intercepts) → strict (additionally equal residual variances). The identification constraints were written out explicitly rather than left to software defaults: group 1 has latent means 0, variances 1 and scale factors 1; equal loadings free the latent variances of group 2; equal thresholds free its latent means and scale factors. The strict model, which needs residual variances as parameters, is fitted in the theta parameterisation and compared with the identical scalar model refitted in theta. Decisions rest primarily on $\Delta\text{CFI} \geq -.010$ and $\Delta\text{RMSEA} \leq .015$, with the stricter $-.005/.010$ for unequal group sizes (Chen, 2007); scaled χ^2 difference tests are reported alongside, and, as recommended in the lecture for threshold models, the scalar model is also compared directly with the configural model (Svetina et al., 2020; Wu & Estabrook, 2016). To locate non-invariance, each item's thresholds were released in turn from the scalar model. Gender is the required grouping variable; age (median split) is a bonus analysis.

3.11 Polytomous IRT

The original responses were analysed with Samejima's graded response model (GRM; Samejima, 1969), in which the cumulative category probabilities are 2PL functions and category probabilities are their differences. It was preferred because the "how much did this apply" judgement is naturally cumulative, and because its normal-ogive form is the polychoric factor model of the CFA, so the polytomous IRT and

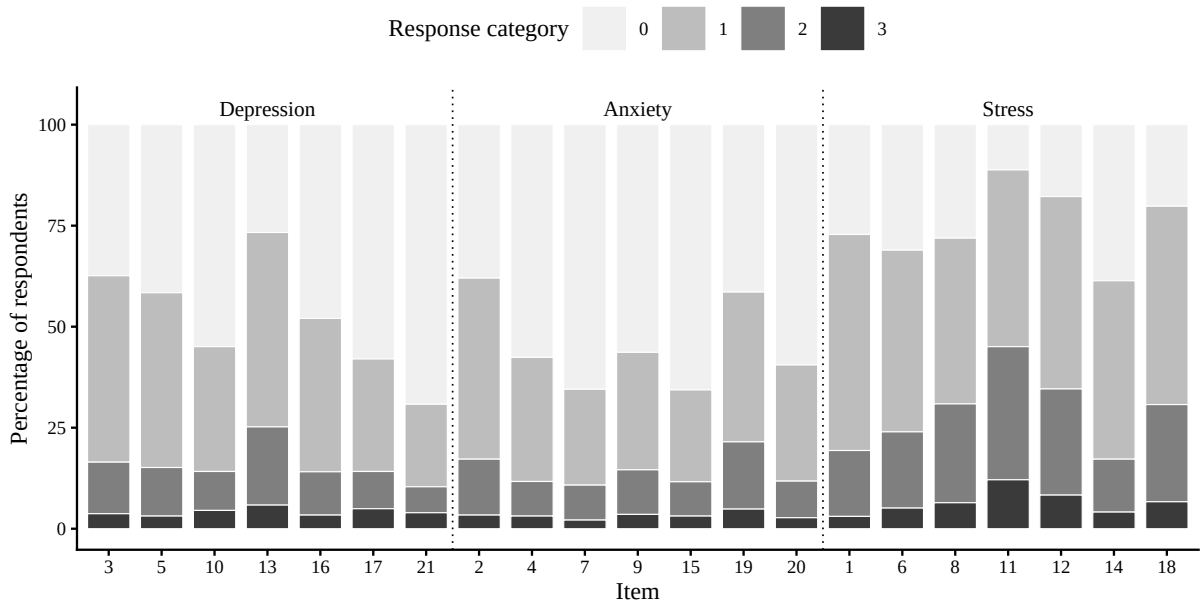


Figure 1: Response distributions of the 21 items, grouped by subscale.

the CFA are two views of one model. For comparison I fitted the GRM with a common slope and the divide-by-total models – partial credit (Masters, 1982) and generalised partial credit (Muraki, 1992); the rating scale model (Andrich, 1978) and the conditions under which sequential and graded formulations coincide (Bechger & Akkermans, 2001) complete the family discussed in the lecture. In the GRM the category boundaries are ordered by construction, so whether the categories function as intended was examined in the PCM/GPCM (ordering of the step parameters) and, for all three models, by checking that every category is the most probable response somewhere on the continuum.

3.12 Software

All analyses were run in R (R Core Team, 2024) as a sequence of R Markdown reports, using `readxl`, `psych`, `eRm`, `mirt`, `difR`, `lavaan`, `semTools`, `BifactorIndicesCalculator` and `ggplot2` (Jorgensen et al., 2022). The reports write every table and figure, and a file of LaTeX macros from which the numbers quoted in this text are taken, so the manuscript can be regenerated from the data.

4 Results

4.1 Descriptive statistics

Table 1 and Figure 1 summarise the items. All are positively skewed, as expected in a general-population sample: means range from 0.45 (item 21, “life was meaningless”) to 1.46 (item 11, “getting agitated”). The floor is substantial for Depression (12.9% score zero) and Anxiety (16.8%) but not for Stress (3.8%) or the total score (2.1%); ceilings are negligible. The total score has $M = 17.15$ ($SD = 11.77$) on a 0–63 range. This already anticipates a recurring result: the scale separates people with elevated symptoms better than people with few, which CTT registers only as skewness and IRT as an information function that falls away at the low end (Figure 4).

4.2 Classical test theory

Internal consistency is high at every level (Table 2): $\alpha = .946$ [.941, .950] for the total scale and .864–.899 for the subscales. Corrected item–total correlations range from .49 (item 2, “dryness of the mouth”) to .76

Table 1: Item descriptives (N = 1214). D/A/S = Depression/Anxiety/Stress; r_{tot}, r_{sub} = corrected item–total correlation with the total scale and with the item's subscale; p_A, p_B = endorsement rates under rules A and B.

Item	Sub	M	SD	% in category				r_{tot}	r_{sub}	p_A	p_B
				0	1	2	3				
3	D	0.83	0.79	37	46	13	4	.68	.66	.63	.16
5	D	0.77	0.78	42	43	12	3	.62	.56	.58	.15
10	D	0.64	0.83	55	31	10	5	.68	.76	.45	.14
13	D	1.04	0.83	27	48	19	6	.76	.75	.73	.25
16	D	0.70	0.79	48	38	11	3	.69	.73	.52	.14
17	D	0.61	0.85	58	28	9	5	.67	.73	.42	.14
21	D	0.45	0.78	69	20	6	4	.65	.72	.31	.10
2	A	0.83	0.79	38	45	14	3	.49	.49	.62	.17
4	A	0.57	0.78	58	31	9	3	.61	.67	.42	.12
7	A	0.47	0.74	65	24	9	2	.60	.65	.35	.11
9	A	0.62	0.82	56	29	11	4	.67	.66	.44	.15
15	A	0.49	0.78	66	23	8	3	.64	.69	.34	.12
19	A	0.85	0.87	41	37	17	5	.60	.62	.59	.21
20	A	0.55	0.77	59	29	9	3	.66	.67	.41	.12
1	S	0.95	0.74	27	53	16	3	.67	.67	.73	.19
6	S	0.98	0.84	31	45	19	5	.65	.67	.69	.24
8	S	1.09	0.88	28	41	24	6	.69	.71	.72	.31
11	S	1.46	0.85	11	44	33	12	.69	.73	.89	.45
12	S	1.25	0.84	18	48	26	8	.71	.72	.82	.35
14	S	0.83	0.81	39	44	13	4	.67	.63	.61	.17
18	S	1.17	0.82	20	49	24	7	.64	.68	.80	.31

Table 2: CTT summary. $\bar{r}$ = mean inter-item correlation; ω = congeneric reliability from the CFA (latent-response metric); ω_{GY} = Green–Yang ω (observed-score metric); SEM = standard error of measurement.

Scale	k	M	SD	Skew	$\bar{r}$	α [95% CI]	ω	ω_{GY}	SEM
Total	21	17.15	11.77	0.98	.45	.946 [.941, .950]	.964	.949	2.75
Depression	7	5.03	4.46	1.26	.56	.899 [.890, .907]	.935	.901	1.42
Anxiety	7	4.38	4.12	1.25	.48	.864 [.852, .875]	.911	.870	1.52
Stress	7	7.73	4.50	0.53	.54	.891 [.881, .900]	.923	.894	1.49

(item 13, "down-hearted and blue"), and no item would raise α if removed.

What does this license? Not the claim that the DASS-21 measures one thing well. α depends only on the number of items and their average correlation ($\bar{r}$ = .45), so with 21 items a modest average correlation is enough for a large coefficient, whether the items share one factor or several correlated ones. It also equals reliability only under essential τ-equivalence, which is untenable here: loadings in the one-factor CFA range from .56 to .84. What α does support – conservatively, since it is a lower bound – is that the total score's standard error of measurement is small (SEM = 2.75 points on the 0–63 scale).

The subscale correlations raise a different question. The sums correlate at r = .68 (D–A), .75 (D–S) and .70 (A–S), and .78, .84 and .80 after correction for attenuation. Constructs that correlate around .8 once measurement error is removed are not "maximally differentiated" – the problem the DASS was designed to solve, and one reported repeatedly for the short form (Antony et al., 1998; Henry & Crawford, 2005; Osman et al., 2012). CTT can raise this question but not answer it, because it cannot separate a general factor from specific ones.

4.3 Binary IRT

Model choice and fit. Under rule A the 2PL fits better than the 1PL, $\chi^2(20) = 83.58$, $p < .001$; AIC prefers the 2PL (24508 vs. 24551), BIC the 1PL (24664 vs. 24722). Andersen's test rejects the Rasch model for a median split of the raw score ($LR = 55.06$, $df = 20$, $p < .001$) and for gender ($LR = 91.12$, $p < .001$). Item-level Rasch fit is better than these global tests suggest: infit lies between 0.84 and 1.14 for every item, inside the conventional 0.7–1.3 band. Outfit is more dispersed (0.57–1.39): item 2 is slightly noisy, and the most frequently endorsed items (11, 12, 13) overfit, which is what one expects when their slopes are steeper than the common Rasch slope. The 2PL fits reasonably by limited-information criteria (M_2-based RMSEA = .065, CFI = .967, SRMSR = .047), and 4 items show $S\text{-}X^2$ misfit at $p < .05$, none after Benjamini–Hochberg correction. My reading is that the rejections reflect real slope heterogeneity, which $N = 1214$ detects easily; the Rasch model is a usable description for the purposes here, but a strict Rasch measurement claim would not survive Andersen's test. I base the substantive interpretation on the 2PL.

Item parameters. Table 3 and Figure 2 give the parameters. Locations range from $b = -1.46$ (item 11, "getting agitated") to $b = 0.65$ (item 21, "life was meaningless"): a severity hierarchy in which everyday stress reactions are endorsed at low trait levels and existential hopelessness only at high ones. Discriminations range from $a = 1.42$ (item 2, dry mouth) to $a = 2.70$ (item 13, down-hearted). The contrast is the clearest illustration of what a slope means: near its location the endorsement probability of item 13 changes far faster with θ than that of item 2, so item 13 separates neighbouring trait levels sharply and item 2 hardly at all. A somatic sign such as a dry mouth is a weak indicator of general negative affect – a substantive finding, not an estimation problem.

Targeting and information. On the Rasch scale the items span -2.65 to 1.84 logits, and the mean person location is 0.54 (Figure 3). Under rule A test information peaks at $\theta = -0.35$ ($I = 15.07$, $SE = 0.26$) and corresponds to a reliability of at least .80 only for $\theta \in [-1.9, 1.2]$; marginal reliability is .897. Under rule B the curve moves to the right: a higher peak (30.95 at $\theta = 1.15$) but a narrower useful range ($[-0.3, 2.4]$) and lower marginal reliability (.813). The cut point therefore does not just rescale the results; it decides *which part of the continuum the binary test measures well*. Rule A measures well among people with mild symptoms, rule B among people with pronounced symptoms and hardly at all below the mean.

Comparison with CTT. The 2PL location and CTT easiness are almost the same information ($r = -.998$). Discrimination agrees much less: under rule A, a correlates only $r = .27$ with the corrected item–total correlation of the binary items, under rule B $r = .84$. The difference is instructive: the point-biserial correlation depends on item difficulty as well as on discrimination, so when difficulties are spread widely (rule A) it no longer tracks a; when they are concentrated (rule B) it does. Person estimates and binary sum scores correlate $r = .988$, so IRT does not *reorder* respondents here; what it adds is an account of how precisely each one is located, which varies across θ and which CTT cannot provide.

4.4 Differential item functioning

Gender, rule A. After Benjamini–Hochberg correction the logistic-regression omnibus test flags 10 items (2, 4, 8, 14, 15, 16, 17, 19, 20, 21), Mantel–Haenszel 8 (2, 4, 14, 15, 16, 17, 19, 20) and Lord's test 0 (full results in Table 11, appendix; Figure 5). The effects run in *both* directions. At the same total score, Group 2 endorses items 4, 15, 17, 19, 20 more often: breathing difficulty, being close to panic, awareness of the heart, feeling scared without reason, and feeling worthless. Group 1 endorses items 2, 8, 14, 16, 21 more often: dry mouth, nervous energy, being intolerant of interruptions, being unable to become enthusiastic, and (only on the omnibus test, with a negligible effect) life feeling meaningless. No item shows non-uniform DIF.

Table 3: Item parameters under rule A (0 vs. 1–3). b_{CML} = Rasch location (conditional ML); infit/outfit = mean-square fit; a, b = 2PL discrimination and location with standard errors.

Item	Sub	p	b_{CML}	SE	Infit	Outfit	a	SE	b	SE
3	D	.63	−0.27	0.07	0.96	0.99	1.96	0.14	−0.43	0.05
5	D	.58	0.01	0.07	1.07	1.11	1.63	0.12	−0.31	0.05
10	D	.45	0.87	0.07	1.00	1.02	1.88	0.13	0.16	0.05
13	D	.73	−1.05	0.08	0.84	0.63	2.70	0.21	−0.74	0.05
16	D	.52	0.42	0.07	0.94	0.88	2.12	0.15	−0.07	0.05
17	D	.42	1.06	0.07	1.00	1.07	1.89	0.13	0.27	0.05
21	D	.31	1.84	0.08	0.96	1.02	2.01	0.15	0.65	0.05
2	A	.62	−0.23	0.07	1.14	1.39	1.42	0.11	−0.48	0.06
4	A	.42	1.04	0.07	1.01	1.05	1.80	0.13	0.26	0.05
7	A	.35	1.57	0.08	1.00	0.90	1.86	0.13	0.54	0.05
9	A	.44	0.96	0.07	0.89	0.84	2.28	0.16	0.19	0.05
15	A	.34	1.58	0.08	1.00	1.09	1.87	0.13	0.54	0.05
19	A	.59	−0.00	0.07	1.07	1.19	1.62	0.12	−0.31	0.05
20	A	.41	1.16	0.07	0.93	0.85	2.11	0.15	0.30	0.05
1	S	.73	−1.01	0.08	0.98	1.11	1.88	0.14	−0.82	0.06
6	S	.69	−0.71	0.08	0.98	0.93	1.90	0.14	−0.66	0.06
8	S	.72	−0.94	0.08	0.96	0.96	1.96	0.14	−0.77	0.06
11	S	.89	−2.65	0.11	0.86	0.57	2.64	0.25	−1.46	0.07
12	S	.82	−1.84	0.09	0.89	0.66	2.43	0.20	−1.13	0.06
14	S	.61	−0.19	0.07	0.96	0.97	2.02	0.14	−0.38	0.05
18	S	.80	−1.61	0.09	1.02	0.95	1.86	0.15	−1.14	0.07

Is any of this practically important? The effect sizes say no. 15 items are ETS class A, 5 class B and 1 class C (item 15, $\Delta_{MH} = 1.74$); the largest ΔR^2 is .014, far below the .035 needed for even a moderate effect, so all 21 items are negligible on the Zumbo–Thomas criterion. Two further points weaken any claim of bias. Because the effects point in opposite directions they largely cancel in the total score. And matching on a total that contains the studied items forces DIF to balance to some extent, so part of the two-sided pattern may be methodological. The content reading – panic and cardiorespiratory symptoms more readily reported by Group 2, low-arousal tension and anhedonia by Group 1 – is a hypothesis about symptom expression or reporting style, not something these data can establish. I retain all items and flag them; deleting items on statistical significance alone at $N > 1200$ would be exactly the error the lecture warns against (Martinková et al., 2017).

Sensitivity to the cut point. Under rule B, 7 items are flagged (11, 12, 13, 17, 18, 19, 20) and 6 reach class C (11, 12, 13, 15, 19, 20); only items 17, 19, 20 are flagged under both rules. DIF conclusions are therefore *not* invariant to the dichotomisation: collapsing categories differently changes which region of the continuum each binary item is sensitive to, and with it where group differences in endorsement become visible.

Bonus: age and education (exploratory). Age produces stronger DIF than gender: 7 items are significant, class C effects appear for items 7, 9, 11, 16, and the largest ΔR^2 is .030. Younger respondents endorse trembling, worry about panicking in public and agitation (items 7, 9, 11) more at equal total score, older respondents the inability to become enthusiastic (item 16) – social-evaluative and arousal symptoms as markers of distress in the young, anhedonia in the old. With age as a continuous covariate 10 items are flagged, including all items flagged by the median split, so the result is not an artefact of the split. Education shows no DIF at all (0 items with the median split, 0 as a continuous covariate, largest $\Delta R^2 = .006$). This null result matters: it shows that the procedures do not flag items indiscriminately at

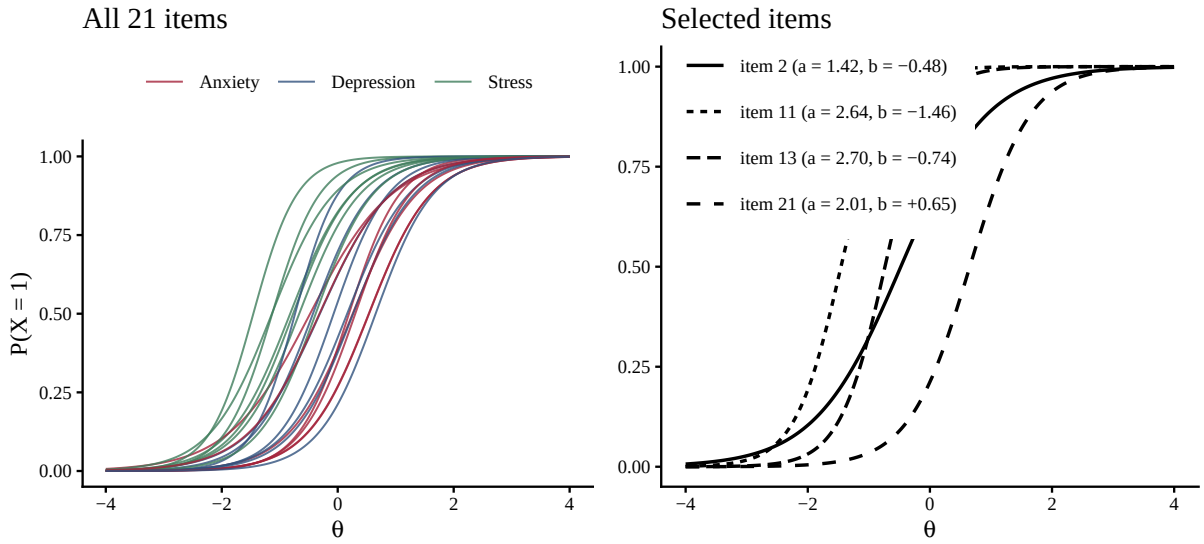


Figure 2: 2PL item characteristic curves, rule A. Left: all items by subscale. Right: four items spanning the range of locations and slopes.

this sample size.

4.5 Confirmatory factor analysis

Table 4 reports the four models. The one-factor model is clearly rejected (CFI = .924, TLI = .916, RMSEA = .104, SRMR = .065). The correlated three-factor model is acceptable by CFI/TLI and SRMR (.970, .966, .040) but its RMSEA of .066 is above the .06 benchmark. The bifactor model fits best (CFI = .980, RMSEA = .057, SRMR = .032), but even it does not reach close fit on every index, and lavaan's robust indices are less favourable still (bifactor robust CFI = .937, RMSEA = .080), although they rank the models in the same order. No model reproduces the polychoric correlations exactly; the comparison is between degrees of approximation. M3 reproduces M2's fit, as it must; its second-order loadings are .91 (Depression), .87 (Anxiety) and .94 (Stress), so nearly all reliable variance in each subscale factor is shared with one higher-order factor.

That the bifactor model fits best is close to uninformative in itself: it has 18 more parameters than M2, and bifactor models win such comparisons almost by construction (Reise, Scheines, et al., 2013). The informative results are elsewhere. *First*, the latent correlations in M2 are very high (φ_{DA} = .79, φ_{DS} = .85, φ_{AS} = .82). *Second*, in M4 the general loadings are uniformly strong (.54–.82) while the specific loadings are weak and uneven (.01–.53; Figure 6): item 14 (λ_S = .01) and item 5 (.07) are pure indicators of the general factor, whereas item 21 (.53), item 10 (.49) and item 4 (.48) – hopelessness and anhedonia on one side, breathlessness on the other – carry the clearest specific variance, exactly the depression/somatic-anxiety contrast the DASS was built on. *Third*, local dependence largely disappears in M4 (Figure 7): the largest residual correlations of M1 are between items 4 and 15 (.17), 19 and 20 (.16) and 4 and 10 (-.15), whereas no residual of M4 exceeds .09 in absolute value. The specific factors thus mostly absorb overlap between near-paraphrases within a subscale (breathlessness and panic; a racing heart and unexplained fear); whether that is a substantive dimension or shared wording is a judgement the model cannot make.

4.6 Reliability, dimensionality and the comparison of ω estimates

Table 5 collects the indices; they answer the two questions of the assignment.

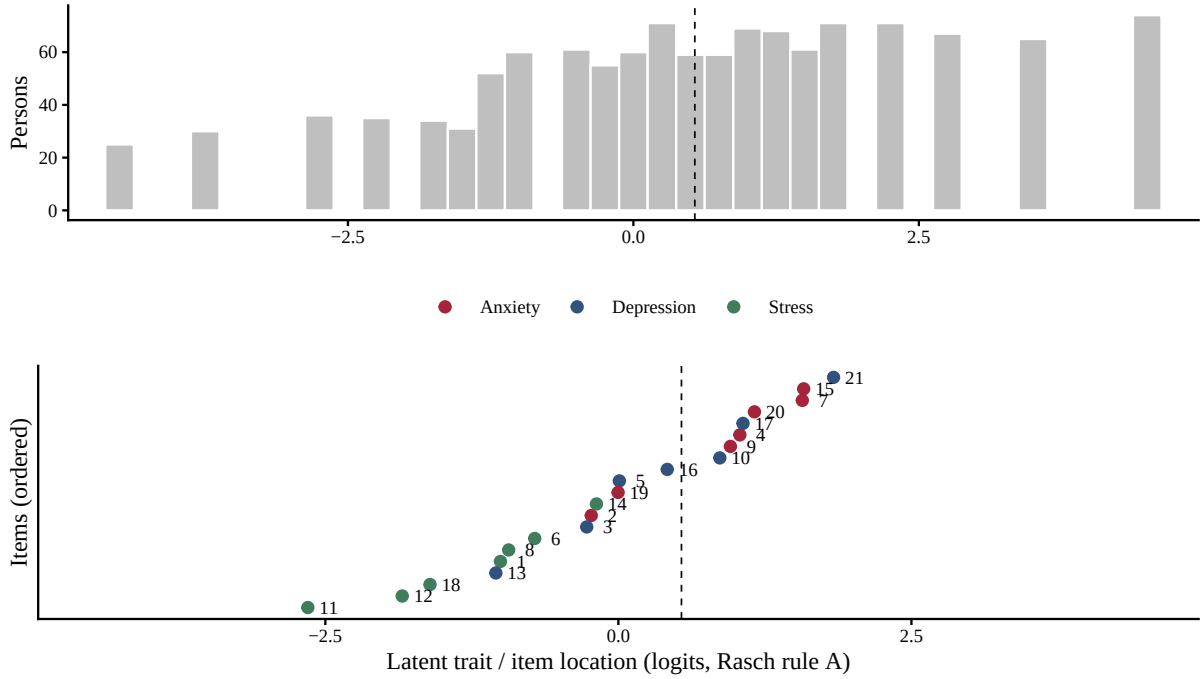


Figure 3: Person–item map, Rasch model, rule A. Top: person locations; bottom: item locations. The dashed line marks the mean person location.

Table 4: Fit of the measurement models (WLSMV, $N = 1214$; scaled indices). [a]Statistically equivalent to M2 with three first-order factors.

Model	q	χ^2	df	CFI	TLI	RMSEA [90% CI]	SRMR
M1. Unidimensional	84	2663.9	189	.924	.916	.104 [.100, .107]	.065
M2. Correlated three-factor	87	1168.6	186	.970	.966	.066 [.062, .070]	.040
M3. Second-order[a]	87	1168.6	186	.970	.966	.066 [.062, .070]	.040
M4. Bifactor	105	823.9	168	.980	.975	.057 [.053, .061]	.032

Can the total score be used? Yes. The general factor accounts for $\omega_H = .908$ of total-score variance against $\omega_T = .970$ for all factors, so .936 of the reliable variance in the total score is due to the general factor. ECV = .808 exceeds the "good" benchmark of .70, construct replicability is $H = .960$ and factor determinacy .962. PUC = .70 is below .80 – with 21 items in three blocks of seven it cannot be higher – which by the lecture's rule means that bias from treating the scale as unidimensional is *not* automatically small and must be judged from the other indices; and these are unambiguous. The total score is a defensible measure of one general dimension of negative affect.

Can the subscale scores be used? Much less so. Each subscale is highly reliable on its own ($\omega = .935$, .911, .923), but almost all of that reliability is borrowed from the general factor: once it is partialled out, the unique reliability is $\omega_S = .169$ (Depression), .224 (Anxiety) and .081 (Stress). Factor determinacies of the specific factors (.815, .796, .737) are below the .80–.90 usually required before factor scores are interpreted, and construct replicability is low ($H = .561$, .564, .391). The contrast for Stress – $\omega = .923$ but $\omega_S = .081$ – is the single most consequential number in the analysis, and one a conventional CTT report never produces. A Stress score is a very reliable measurement – of general negative affect; as a measure of something specific to stress it retains less than a tenth of its variance. Difference scores between subscales, or models entering all three as separate predictors, are poorly supported.

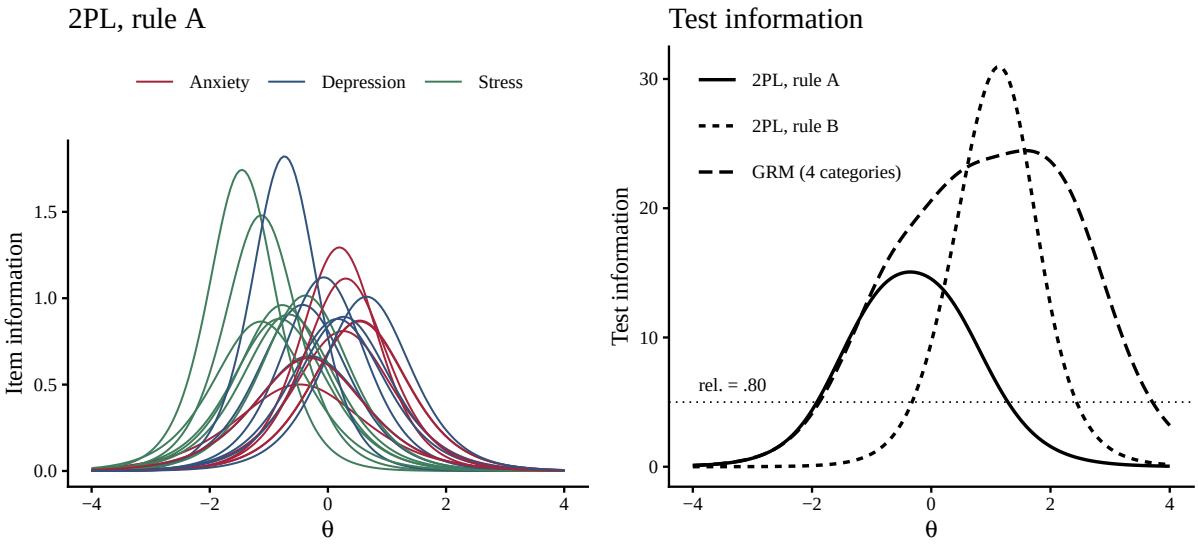


Figure 4: Left: item information, 2PL rule A. Right: test information of the graded response model and of the two dichotomised 2PL solutions; the dotted line marks the information corresponding to a reliability of .80.

Why the ω estimates differ. Table 6 lists ten reliability estimates for the total score. They differ systematically, and each difference has an interpretation.

1. *Assumed versus estimated loadings.* α = .946 is below the congeneric ω = .964; the gap (.019) is what assuming equal loadings costs.
2. *Latent versus observed metric.* Ordinal α (.962) and the linear ω coefficients describe a sum of the continuous latent responses Y^*, a score nobody observes. The Green–Yang coefficients describe the observed 0–3 sum and are lower (ω: .949 vs. .964; ω_H: .888 vs. .908). As the lecture puts it, the size of this gap (here .016 and .020) estimates how far the linearity and normality assumptions are violated – modest, given how skewed the items are. `BifactorIndicesCalculator`, given the ordinal model, returns exactly the Green–Yang values (ω = .955, ω_H = .888), an independent confirmation of the hand computation.
3. *All common factors versus the general factor.* ω_T includes the specific factors, ω_H does not. Reporting only ω_T for a multidimensional scale invites attributing all reliable variance to the construct in the scale's title.
4. *Confirmatory versus exploratory general factor.* The Schmid–Leiman ω_H (.862) is lower than the confirmatory one (.908) while ω_T is the same (.969 vs. .970). The confirmatory model forces each item onto one specific factor and lets the general factor absorb everything else; the exploratory solution derives the general factor from the correlations of oblique group factors and allows cross-loadings, assigning more variance to the group factors. The "true" ω_H is therefore not a single number but depends on how the general factor is defined – the reason the lecture warns that the functions "can differ".

Every coefficient in Table 6 is correct; they answer different questions about different scores. For the observed total score, the defensible statements are $\omega_{GY} \approx .949$ for reliability and $\omega_H \approx .888$–.908 for the part attributable to the general factor.

4.7 Measurement invariance

Table 7 reports the sequence for the three-factor and the one-factor model. The configural three-factor model fits acceptably in both groups (CFI = .971, RMSEA = .065), so the same factor structure holds and the comparison is meaningful.

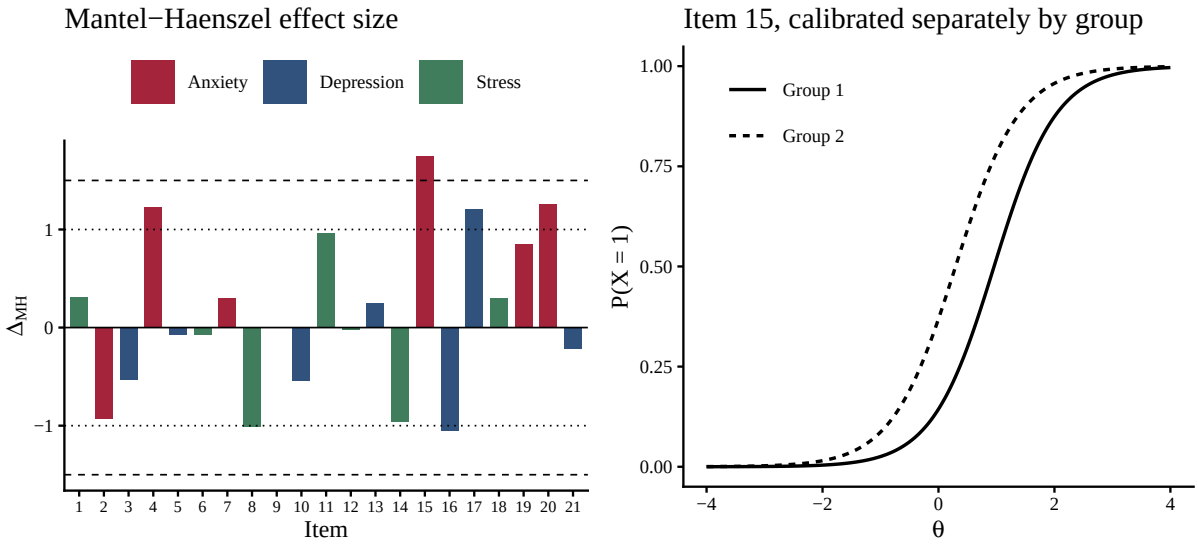


Figure 5: Left: Mantel–Haenszel effect sizes by item (gender, rule A); dotted and dashed lines mark $|\Delta_{MH}| = 1$ and 1.5. Right: item characteristic curves of the item with the largest effect, calibrated separately in each group.

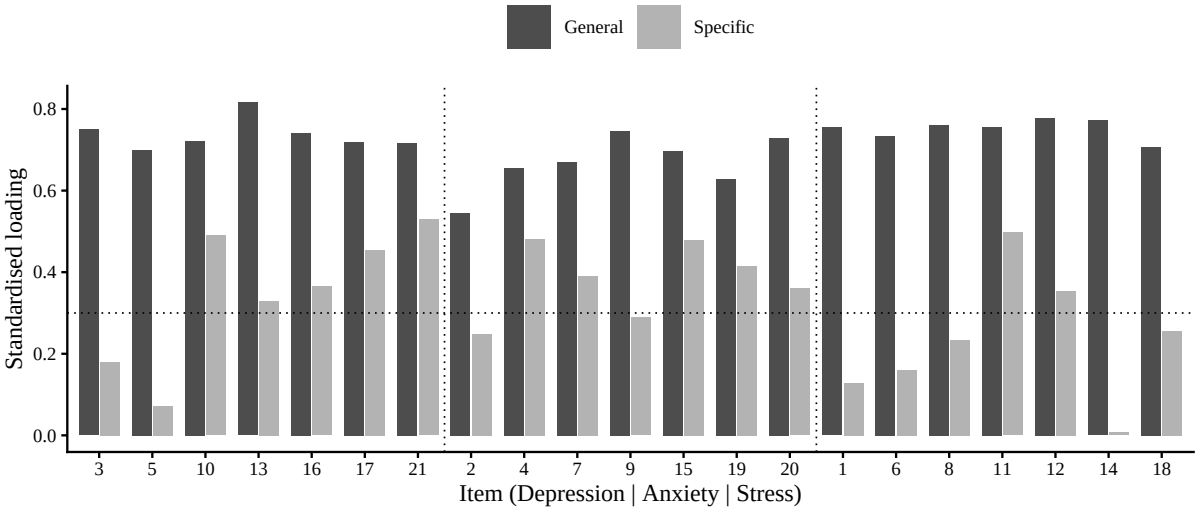


Figure 6: Standardised loadings in the bifactor model. Dotted line: $\lambda = .30$.

Loadings. Constraining them is supported by the difference test, $\Delta\chi^2(18) = 27.7$, $p = .068$, and does not worsen approximate fit. Metric invariance holds, consistent with the absence of non-uniform DIF: the items relate to the factors equally strongly in both groups.

Thresholds. Here the criteria diverge. Relative to the metric model, $\Delta\chi^2(39) = 58.8$ ($p = .022$) and ΔCFI = −.008, ΔRMSEA = +.007 – inside the general −.010 criterion but outside Chen's stricter −.005 for unequal groups. These consecutive comparisons are, however, distorted by the scaled-and-shifted statistic, which is not monotone across nested models: the metric model has a *better* CFI (.980) than the less constrained configural model (.971). This is precisely why the lecture recommends comparing the threshold model directly with the configural model. That comparison gives ΔCFI = +.000 and ΔRMSEA = −.005 – no deterioration – while the difference test, $\Delta\chi^2(57) = 134.6$, $p < .001$, rejects exact equality. I conclude that thresholds are *approximately* invariant: exact invariance is rejected, but the misfit it causes is small. Equal residual variances add no meaningful misfit ($\Delta\chi^2(21) = 30.5$, $p = .083$; ΔCFI = +.007).

Where is the non-invariance? Releasing each item's thresholds in turn, 10 items improve fit significantly after correction, most of all item 17 ($\Delta\chi^2 = 24.5$), followed by items 17, 19, 2, 15. Items 17, 19 and 15

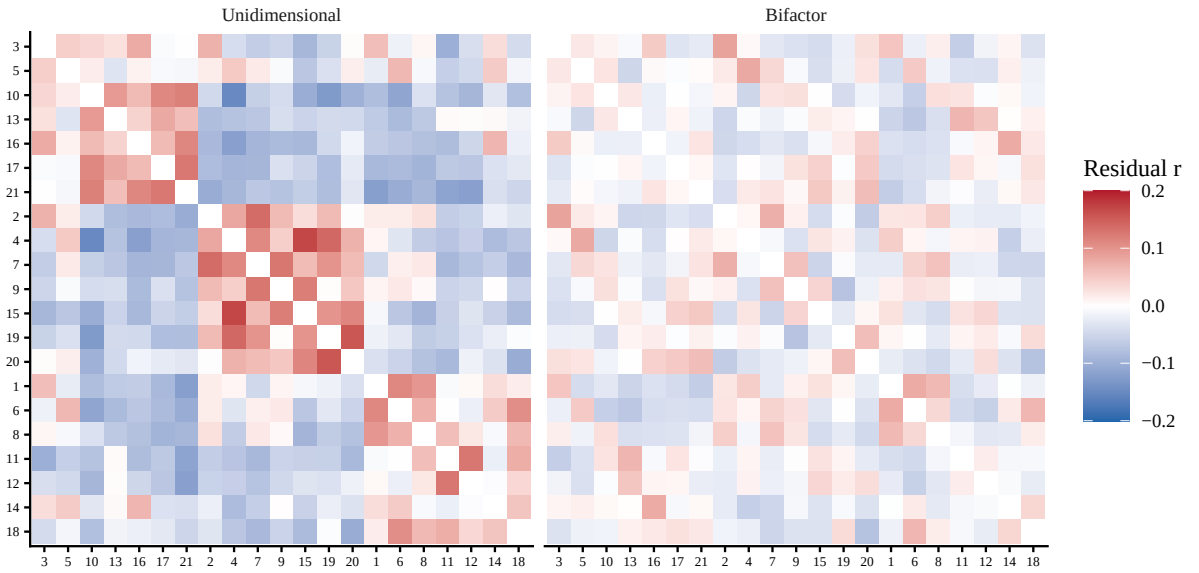


Figure 7: Residual polychoric correlations of the one-factor and the bifactor model, items ordered by subscale. The within-subscale blocks visible on the left are what the specific factors absorb.

are also flagged in the DIF analysis, and item 2 is among the Group-1-favoured DIF items: two different procedures, applied to binary versus four-category data, point at the same items. Freeing the thresholds of the four worst items changes the latent mean differences only slightly (Depression 0.25 → 0.22, Anxiety 0.39 → 0.33, Stress 0.30 → 0.30 SD).

What may be compared? The factor structure and the loadings are comparable across groups, and latent means can be compared with the caveat of approximate threshold invariance: Group 2 is higher by 0.25 (Depression), 0.39 (Anxiety) and 0.30 (Stress) SD, and by 0.32 SD on the single general factor. Whether *observed sum scores* may be compared is a separate question, since sum-score bias depends on whether threshold differences cancel or accumulate across items. Here they largely cancel: the standardised differences of the observed subscale sums (.22, .34, .28) are close to the latent differences under both the full and the partial model. The one-factor model shows the same pattern but its configural fit is poor (CFI = .928, RMSEA = .101), so its invariance tests describe a model that does not fit well to begin with.

Bonus: age (exploratory). For the age split (Table 12, appendix) loadings are again invariant ($p = .616$), but the threshold step fails both consecutive criteria (Metric → scalar: ΔCFI = −.011, $p < .001$) and strict invariance is rejected by the difference test ($p = .002$); only the direct configural–scalar comparison (ΔCFI = −.001) is unproblematic. Older respondents score lower on all three factors (-0.18, -0.31, -0.30 SD). Together with the stronger age DIF, this means comparisons between age groups rest on a weaker measurement basis than comparisons between the gender groups.

4.8 Polytomous IRT and the cost of dichotomisation

The graded response model is the best of the polytomous models by AIC and BIC (Table 8); it fits clearly better than the GPCM with the same number of parameters and than the GRM with one common slope ($\Delta\chi^2(20) = 214.7$). The items therefore differ in how sharply they separate trait levels, and the cumulative formulation suits these data better than the adjacent-category one. Discriminations range from 1.24 (item 2) to 2.87 (item 13) – the same extremes as in the binary analysis – with three ordered boundaries per item (Table 9).

Table 5: Reliability and dimensionality indices (bifactor model unless stated).

Index	Value	Interpretation
ω, unidimensional model (linear)	.964	sum-score variance due to one common factor (latent-response metric)
ω, unidimensional model (Green–Yang)	.949	same, in the metric of the observed 0–3 scores
ω_T, bifactor	.970	variance due to general *and* specific factors
ω_H, bifactor (linear)	.908	variance due to the general factor alone
ω_H, bifactor (Green–Yang)	.888	same, observed-score metric
ω_H/ω_T	.936	share of reliable variance due to the general factor
ECV	.808	share of common variance explained by the general factor
PUC	.70	share of item correlations not affected by the specific factors
Factor determinacy, general	.962	correlation of factor-score estimates with the factor
Depression: ω / ω_S / FD	.937 / .169 / .815	subscale total, unique part, determinacy
Anxiety: ω / ω_S / FD	.912 / .224 / .796	subscale total, unique part, determinacy
Stress: ω / ω_S / FD	.924 / .081 / .737	subscale total, unique part, determinacy

Table 6: Ten reliability estimates for the DASS-21 total score.

Estimate	Metric	Value
Cronbach's α (raw item scores)	observed score	.946
Ordinal α (polychoric correlations)	latent response	.962
ω, unidimensional CFA (linear)	latent response	.964
ω, unidimensional CFA (Green–Yang)	observed score	.949
ω_T, bifactor CFA (linear)	latent response	.970
ω_T, bifactor CFA (Green–Yang)	observed score	.955
ω_H, bifactor CFA (linear)	latent response	.908
ω_H, bifactor CFA (Green–Yang)	observed score	.888
ω_H, EFA + Schmid–Leiman	latent response	.862
ω_T, EFA + Schmid–Leiman	latent response	.969

Category functioning. Because the GRM orders its boundaries by construction, it cannot show disordered thresholds; the meaningful check is in the divide-by-total models. In both the PCM and the GPCM all step parameters are ordered (0 and 0 items with disordered steps), and in all three models every category is the most probable response somewhere on the continuum (Figure 8). Respondents use the four categories in the intended order, so there is no empirical reason to collapse them.

Fit. By limited-information criteria the unidimensional GRM fits only approximately (M_2-based RMSEA = .099, CFI = .958, SRMSR = .059), while no item shows $S\text{-}X^2$ misfit after correction. This mirrors the CFA: the misfit is not located in single items but in the residual correlations within subscales that a one-dimensional model cannot reproduce.

Information. The GRM reaches its maximum information (24.46) at $\theta = 1.60$ and provides a reliability of at least .80 over $\theta \in [-1.9, 3.7]$; its marginal reliability is .941, against .897 (rule A) and .813 (rule B). Mean standard errors of the person estimates are 0.24, 0.31 and 0.39 respectively: under rule B a respondent is located with an error 1.65 times as large. Integrated over the continuum the GRM carries 2.43 times the information of the rule-A 2PL and 1.84 times that of rule B (Figure 4). This ratio should be read as precision per unit of a standardised trait, since each model standardises its own θ; the marginal

Table 7: Measurement invariance across gender (WLSMV, four-category data, scaled indices). Metric = equal loadings; scalar = equal loadings and thresholds; strict = additionally equal residual variances. Δ columns compare each model with the previous one; the last row of each block compares scalar with configural directly.

Model	Step	χ^2	df	CFI	RMSEA	$\Delta\chi^2$ (df)	p	ΔCFI	ΔRMSEA
Three-factor	Configural	1315.4	372	.971	.065				
	Metric	1046.9	390	.980	.053	27.7 (18)	.068	+.009	−.012
	Scalar	1359.9	429	.971	.060	58.8 (39)	.022	−.008	+.007
	Strict	1167.4	450	.978	.051	30.5 (21)	.083	+.007	−.009
	Scalar vs. configural					134.6 (57)	< .001	+.000	−.005
One-factor	Configural	2695.6	378	.928	.101				
	Metric	1875.3	398	.954	.078	28.3 (20)	.104	+.026	−.022
	Scalar	2539.3	439	.935	.089	53.0 (41)	.099	−.019	+.011
	Strict	2000.8	460	.952	.074	31.1 (21)	.072	+.017	−.015
	Scalar vs. configural					130.7 (61)	< .001	+.007	−.012

Table 8: Polytomous IRT models. q = number of parameters.

Model	q	$\log L$	AIC	BIC
GRM	84	−22019.6	44207	44636
GRM, common slope	64	−22126.9	44382	44708
GPCM	84	−22063.9	44296	44724
PCM	64	−22183.8	44496	44822

reliabilities make the same point in a conventional metric. Rule B is instructive: its *peak* information (30.95) is higher than the GRM's, but over a narrow band at the top of the scale. Dichotomisation does not lose information uniformly; it concentrates what remains around the cut point.

GRM person estimates correlate .916 and .859 with the rule-A and rule-B estimates and .977 with the raw sum: dichotomisation largely preserves the ranking of respondents and costs precision, especially in the tails. The exercise required a binary analysis, and it is useful for making the item parameters and the Rasch fit machinery visible in their simplest form; for measuring individuals with this instrument, however, the four categories should be kept.

5 Integrated Discussion

All analyses were run on the same respondents so that they can be compared directly.

5.1 Where the methods agree

Ordering of persons. Sum scores, 2PL and GRM person estimates rank respondents almost identically ($r = .988$ between the 2PL estimate and the binary sum, $r = .977$ between the GRM estimate and the raw sum). If the only aim is to order people on one continuum, the sophisticated methods do not do it better. What IRT adds is an account of the *precision* of that ordering, of comparability across groups, and of the behaviour of individual items.

Ordering and quality of items. CTT easiness and IRT location carry the same information ($r = -.998$). The CFA loadings, transformed into logistic discriminations by $a_i = 1.702\,\lambda_i/\sqrt{1-\lambda_i^2}$ (lecture 6), correlate .84 with the 2PL slopes estimated directly from the binary data (Figure 9, appendix). Since one set comes from four-category polychoric correlations and the other from binary responses by marginal maximum likelihood, this is a concrete demonstration that item factor analysis and IRT are one model in two notations; the remaining scatter reflects the information lost by dichotomising. item 2 (dry mouth)

Table 9: Graded response model parameters. b_1–b_3 = category boundaries; spread = $b_3 - b_1$; $I_{\max}$ = maximum item information.

Item	Sub	a	b_1	b_2	b_3	Spread	$I_{\max}$
3	D	2.16	−0.41	1.19	2.37	2.78	1.28
5	D	1.73	−0.31	1.39	2.73	3.04	0.84
10	D	2.27	0.13	1.29	2.17	2.04	1.50
13	D	2.87	−0.71	0.74	1.85	2.56	2.16
16	D	2.25	−0.08	1.31	2.39	2.46	1.41
17	D	2.18	0.23	1.33	2.17	1.94	1.40
21	D	2.26	0.59	1.55	2.28	1.68	1.53
2	A	1.24	−0.52	1.56	3.24	3.76	0.44
4	A	1.71	0.24	1.64	2.75	2.51	0.86
7	A	1.71	0.53	1.71	3.03	2.51	0.86
9	A	2.12	0.18	1.29	2.37	2.20	1.29
15	A	1.95	0.49	1.53	2.58	2.08	1.12
19	A	1.58	−0.34	1.10	2.51	2.85	0.72
20	A	2.03	0.28	1.49	2.63	2.35	1.17
1	S	2.03	−0.79	1.08	2.56	3.35	1.10
6	S	1.92	−0.66	0.91	2.24	2.90	1.02
8	S	2.12	−0.74	0.61	1.99	2.73	1.23
11	S	2.27	−1.50	0.15	1.46	2.96	1.38
12	S	2.38	−1.12	0.47	1.71	2.83	1.51
14	S	2.06	−0.38	1.19	2.32	2.70	1.18
18	S	1.87	−1.12	0.65	2.06	3.18	0.96

is the weakest item and item 13 (down-hearted) the strongest on every criterion: corrected item–total correlation, CFA loading, 2PL slope and GRM slope.

Dimensionality. The disattenuated subscale correlations (.78–.84), the latent correlations (.79–.85), the second-order loadings (.87–.94) and the bifactor indices (ECV = .808, ω_H/ω_T = .936) say the same thing from four directions: one strong general dimension with thin specific residues. Earlier Rasch analyses of the DASS-21 likewise found that the full scale does not meet strict Rasch requirements as one measure (Shea et al., 2009); the present data agree that a single *strict* unidimensional model is too restrictive while a general factor clearly dominates.

Gender. DIF and invariance testing agree that loadings/slopes are equal (no non-uniform DIF; metric invariance holds), that a few items differ in location (uniform DIF; threshold non-invariance), that these are largely the same items, and that the differences are small and partly cancel.

5.2 Where the methods disagree, and why

Discrimination in CTT and IRT. The corrected item–total correlation tracks the 2PL slope poorly under rule A (r = .27) and well under rule B (r = .84). The CTT index confounds discrimination with difficulty, so an applied researcher who deletes the item with the lowest item–total correlation may delete the item with the most extreme location rather than the least informative one.

Is the scale unidimensional? The one-factor CFA is rejected (CFI = .924, RMSEA = .104), whereas the bifactor indices say the scale is essentially unidimensional (ω_H = .908, ECV = .808). Both are right because they answer different questions. The CFA asks whether one factor reproduces the correlations *exactly*, and at N = 1214 it detects the local dependence within subscales. The bifactor indices ask

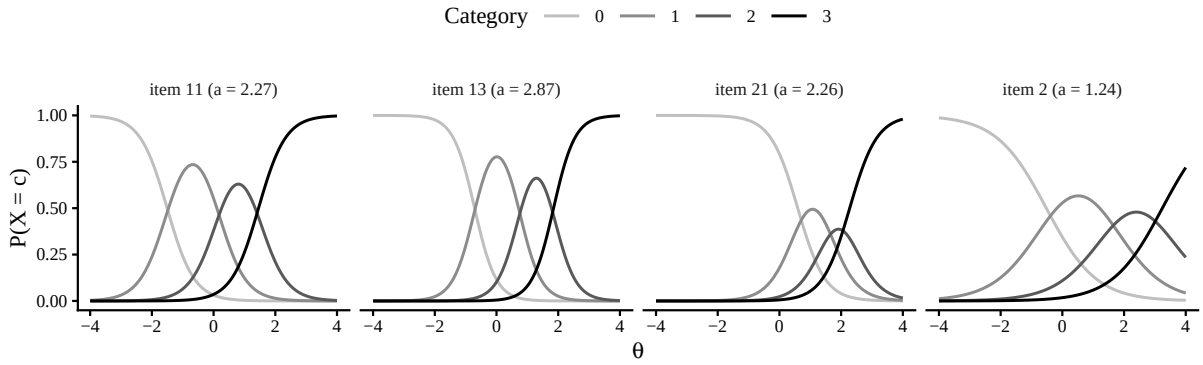


Figure 8: GRM category response curves for four items spanning the range of discriminations. Every category is the most probable response over some region of θ.

how much bias scoring the scale as unidimensional would introduce: little. Statistical and practical dimensionality are different properties, and the bifactor framework is valuable precisely because it separates them.

Are the thresholds invariant? Difference tests reject exact equality, the direct configural–scalar ΔCFI shows no deterioration, consecutive ΔCFI values are inflated or deflated by the non-monotone scaled statistic, and the DIF effect sizes are negligible. These are not contradictions but answers to different questions: significance tests ask whether differences are exactly zero (they are not), fit and effect size criteria ask whether they are large enough to matter (they are not). Reporting only one of them would misrepresent the data.

Do the binary and polytomous analyses agree? On the ordering of items and persons, yes; on precision, no (Section 4.8); and on DIF only partly, since the flagged items depend on the cut point. Dichotomisation is not a neutral preprocessing step, even for questions that seem far removed from it.

5.3 What each framework contributed

CTT alone would have reported $\alpha = .946$ and three reliable subscales and concluded that the DASS-21 works well – missing that the subscales are barely separable and that precision is lowest where many respondents are. *IRT* revealed targeting, conditional precision and the item hierarchy, but a unidimensional model cannot ask whether three dimensions exist. *CFA* established the structure and the misfit of the one-factor model but says nothing about where on the continuum the scale measures well. *The bifactor model* answered the scoring question none of the others can pose. *DIF and invariance testing* were needed before any group comparison could be interpreted. *The polytomous model* showed what the binary analysis cost. The methods are not competing answers to one question; they answer different questions that a complete evaluation of an instrument has to ask.

6 Limitations

Design. The data are cross-sectional, from one non-clinical population, without an external criterion, clinical diagnosis or retest. Nothing here speaks to criterion validity or test–retest reliability, or to whether the same general factor would emerge in a clinical sample, where symptom profiles rather than overall severity may vary more.

Coding and content. Without a codebook the gender labels are an inference (Section 3.2); the content interpretation of the DIF pattern would reverse if the labels did. The content interpretations also assume

the published English item wording; the data contain no item text.

Model fit. No model fits exactly, and the conclusions rest on approximate fit. The choice between scaled and robust fit indices matters for absolute judgements (the robust indices are less favourable) though not for the ranking of models. Every IRT model fitted assumes local independence, which the residual correlations within subscales mildly violate; this slightly inflates slopes and reliability for the affected items.

Bifactor interpretation. The bifactor model fits better partly because it is more flexible, and bifactor solutions are sample-dependent (Reise, Scheines, et al., 2013); its superior fit is not evidence of a bifactor *process*. The conclusions rest on the loading pattern and the reliability decomposition, and the Schmid–Leiman ω_H (.862) shows that the size of the general factor depends on how it is defined.

DIF and invariance. The DIF procedures match on an observed score that contains the studied items, which can create spurious balancing effects; purification mitigates but does not remove this. The multi-group models are large relative to the smaller group (n = 486), the ΔCFI cut-offs were derived for continuous indicators, and the age and education analyses use median splits (although the continuous-covariate check reproduces the age result). The bonus analyses are exploratory.

What cannot be concluded. These data do not show that the subscales are invalid, only that in this sample their unique reliable variance is small. They do not show that any item is biased in a way that warrants removal, only that some function slightly differently. And they cannot settle whether a general factor of negative affect is a psychological entity or a convenient summary; a factor analysis of one questionnaire has nothing decisive to say about that.

7 Conclusion

In this sample the DASS-21 is a reliable measure of one broad dimension of negative affect. A total score is defensible: the general factor accounts for .936 of its reliable variance (ECV = .808). Separate subscale scores are much harder to justify, because after the general factor is partialled out they retain only ω_S = .081–.224 of unique reliable variance and their factor scores are not determinate enough to interpret – a limit that remains invisible in a conventional CTT report. Measurement across the two gender groups is comparable for practical purposes: loadings are invariant, thresholds approximately so, the items with differential functioning have negligible effects in both directions, and latent mean differences barely change when the least invariant items are freed. Age is a stronger source of differential functioning than gender, education none at all.

The most instructive result is methodological. Dichotomising the four-category items – required by the exercise and common in practice – costs much of the test information, moves the region in which the instrument measures well, and changes which items appear biased. Running the binary analysis under two rules and against the graded model was more informative than running it once.

Use of AI Assistance

As requested in the assignment, I declare that I used an AI assistant (Claude, Anthropic) for this essay: for debugging the R code when I had errors and also for editing and revising the text. I ran all analyses myself, checked the methodological choices against the course material, and am responsible for the interpretations. All numbers in the text and tables are generated by the R code from the data.

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A Supplementary Tables and Figures

Table 10: Item parameters under rule B (0–1 vs. 2–3), sensitivity analysis.

Item	Sub	p	b_{CML}	Infit	Outfit	a (2PL)	b (2PL)
3	D	.16	0.24	0.90	0.90	2.86	1.15
5	D	.15	0.40	1.05	1.12	2.16	1.32
10	D	.14	0.53	0.88	0.79	3.10	1.23
13	D	.25	−0.65	0.83	0.72	3.66	0.77
16	D	.14	0.54	0.91	0.92	2.84	1.26
17	D	.14	0.53	0.87	0.84	3.20	1.22
21	D	.10	1.07	0.90	0.84	2.94	1.45
2	A	.17	0.16	1.35	1.79	1.42	1.48
4	A	.12	0.87	1.02	1.09	2.23	1.50
7	A	.11	1.01	1.00	1.12	2.24	1.56
9	A	.15	0.47	0.94	0.84	2.68	1.26
15	A	.12	0.88	0.90	0.78	2.82	1.39
19	A	.21	−0.30	1.12	1.18	2.06	1.04
20	A	.12	0.86	0.93	0.61	2.91	1.37
1	S	.19	−0.08	0.95	0.85	2.69	1.04
6	S	.24	−0.54	1.04	1.06	2.27	0.91
8	S	.31	−1.16	0.98	0.94	2.60	0.63
11	S	.45	−2.36	0.98	0.87	2.86	0.17
12	S	.35	−1.48	0.92	0.88	3.03	0.49
14	S	.17	0.16	0.99	0.98	2.49	1.16
18	S	.31	−1.15	1.09	1.26	2.20	0.66

Table 11: DIF with respect to gender, rule A. Δ_{MH} = ETS delta (positive: Group 2 endorses more at equal total score); ETS = A/B/C class; LR_u, LR_{nu} = uniform and non-uniform likelihood-ratio statistics; ΔR^2 = Nagelkerke effect size with Zumbo–Thomas class; χ^2_L = Lord's statistic. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$ (unadjusted).

Item	Sub	χ^2_{MH}	Δ_{MH}	ETS	LR_u	LR_{nu}	ΔR^2	Z–T	χ^2_L
3	D	1.74	−0.54	A	2.33	0.35	.002	A	2.18
5	D	0.02	−0.07	A	0.20	1.92	.000	A	2.99
10	D	1.74	−0.54	A	2.34	0.18	.002	A	3.02
13	D	0.23	0.25	A	0.24	1.01	.001	A	1.10
16	D	7.33**	−1.05	B	8.94**	0.43	.007	A	7.03*
17	D	9.35**	1.20	B	9.65**	0.03	.007	A	3.09
21	D	0.21	−0.22	A	0.13	3.16	.008	A	2.60
2	A	6.50*	−0.93	A	7.81**	0.02	.007	A	6.21*
4	A	10.69**	1.23	B	11.85***	0.01	.008	A	4.80
7	A	0.49	0.30	A	0.56	2.14	.002	A	3.92
9	A	0.00	0.01	A	0.04	0.72	.000	A	2.53
15	A	18.84***	1.74	C	19.66***	0.10	.014	A	8.13*
19	A	5.88*	0.85	A	6.41*	1.45	.006	A	3.74
20	A	10.10**	1.26	B	9.22**	0.52	.007	A	2.58
1	S	0.47	0.31	A	0.46	0.59	.003	A	0.59
6	S	0.01	−0.07	A	0.07	0.93	.000	A	1.46
8	S	5.22*	−1.01	B	6.54*	1.52	.006	A	5.01
11	S	2.35	0.96	A	2.17	0.00	.003	A	1.39
12	S	0.00	−0.02	A	0.10	0.37	.001	A	0.20
14	S	5.75*	−0.97	A	7.44**	0.34	.005	A	5.32
18	S	0.37	0.30	A	0.50	2.73	.003	A	2.85

B Reproducing the Analysis

The project folder contains the raw data (folder `data`), ten R Markdown reports numbered 00 to 09 (folder `analysis`), a shared setup file (folder `R`) and this document (folder `essay`). Running the script `run_all.R` from the project folder renders all reports in order. The last report writes the tables, the figures and the macro file `numbers.tex` from which the numbers in the text are taken, so recompiling this document afterwards reproduces the essay from the data.

Table 12: Measurement invariance across age groups (bonus analysis; layout as Table 7).

Model	Step	χ^2	*df*	CFI	RMSEA	$\Delta\chi^2$ (*df*)	*p*	ΔCFI	ΔRMSEA
Three-factor	Configural	1395.8	372	.969	.067				
	Metric	1075.8	390	.980	.054	15.7 (18)	.616	+.010	−.014
	Scalar	1476.8	429	.969	.064	152.2 (39)	< .001	−.011	+.010
	Strict	1293.5	450	.975	.056	44.3 (21)	.002	+.006	−.008
	Scalar vs. configural					161.5 (57)	< .001	−.001	−.004
One-factor	Configural	2822.2	378	.927	.103				
	Metric	1999.5	398	.952	.082	42.8 (20)	.002	+.025	−.022
	Scalar	2696.9	439	.933	.092	15.2 (41)	1.000	−.020	+.011
	Strict	2203.3	460	.948	.079	69.6 (21)	< .001	+.015	−.013
	Scalar vs. configural					158.8 (61)	< .001	+.006	−.011

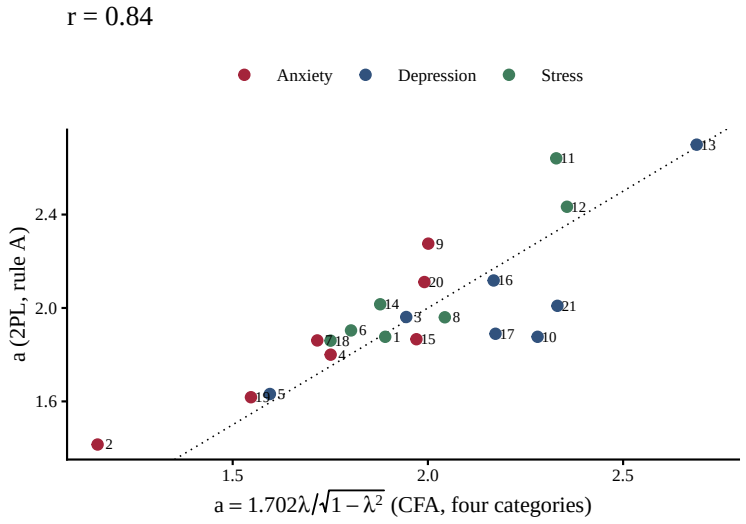


Figure 9: Discriminations implied by the one-factor CFA loadings versus 2PL discriminations estimated from the rule-A data.

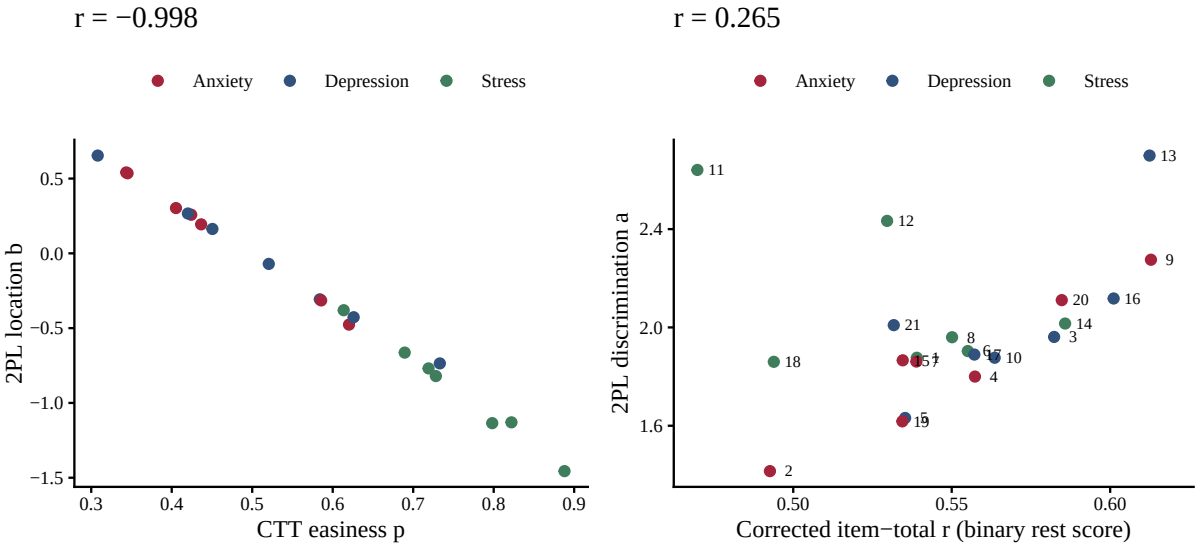


Figure 10: CTT and IRT item indices, rule A.